

SQL Tuning

Functional Dependencies

- Functional dependencies are denoted by σ
- Given: $X \rightarrow Y$.
If two tuples of r (a table with relation schema R) agree on their X-values, then they agree on their Y-values.
- Note: SQL query cannot be used in CHECK constraints
- Functional dependencies may be enforced when LHS has PRIMARY KEY or UNIQUE constraints.
- **Trivial** FD: $X \rightarrow Y$ where $Y \subseteq X$
- **Non-trivial** FD: $X \rightarrow Y$ where $Y \subsetneq X$
- **Completely non-trivial** FD: $X \rightarrow Y$ where $Y \neq \emptyset$ and $Y \cap X = \emptyset$
- Course convention allows: $\emptyset \rightarrow \emptyset, \emptyset \rightarrow Y, X \rightarrow \emptyset$
- **Superkey**: X is a superkey if $X \rightarrow R$ (i.e. X functionally determines all attributes in R)
- **Candidate key**: X is a candidate key if X is a superkey and there is no proper subset of X that is a superkey. i.e. $X \rightarrow R$ and $\nexists Y \subsetneq X$ such that $Y \rightarrow R$, X is minimal.
- **Logical Entailment**: $F \models f$ if every relation that satisfies F also satisfies f . E.g., if $F = \{X \rightarrow Y, Y \rightarrow Z\}$, then $F \models X \rightarrow Z$.

Armstrong's Axioms

- **Reflexivity**: If $Y \subseteq X$, then $X \rightarrow Y$ (i.e. trivial FDs are always valid)
- **Augmentation**: If $X \rightarrow Y$, then $XZ \rightarrow YZ$ for any Z (You can add the same attributes to both sides of an FD and it still holds)
- **Transitivity**: If $X \rightarrow Y$ and $Y \rightarrow Z$, then $X \rightarrow Z$ (i.e. You can short circuit the arrows)
- **Union Rule**: If $X \rightarrow Y$ and $X \rightarrow Z$, then $X \rightarrow YZ$
- **Decomposition Rule**: If $X \rightarrow YZ$, then $X \rightarrow Y$ and $X \rightarrow Z$
- **Pseudotransitivity**: If $X \rightarrow Y$ and $YW \rightarrow Z$, then $XW \rightarrow Z$
- **Sound**: If an FD can be derived from Armstrong's Axioms, then it is logically entailed by the set of FDs.
- **Complete**: If an FD is logically entailed by a set of FDs, then it can be derived from Armstrong's Axioms.

Closure of FDs and Attributes

- The closure of a set of functional dependencies Σ (denoted Σ^+) is the set of all FDs that can be logically entailed by Σ .
- $\Sigma^+ = \{X \rightarrow Y \mid \Sigma \models X \rightarrow Y\}$
- **Attribute Closure**: The closure of a set of attributes X under a set of FDs Σ (denoted X^+) is the set of all attributes that are functionally determined by X under Σ .
- $X^+ = \{A \mid \exists (X \rightarrow \{A\}) \in \Sigma^+\}$
- Algorithm to compute **Attribute Closure**:
 1. Initialize $X^+ = X$
 2. For each FD $Y \rightarrow Z$ in Σ , if $Y \subseteq X^+$, then add Z to X^+
 3. Repeat step 2 until no more attributes can be added to X^+

- **Equivalence of FD sets**: Two sets of FDs Σ and Σ' are equivalent if $\Sigma^+ = \Sigma'^+$ (i.e. they have the same closure).
- To check for equivalence, check if they cover each other:
 1. For each FD in Σ_A , use FDs in Σ_B to compute the attribute closure of the LHS of the FD.
 2. If the closure contains the RHS, then the FD is implied by Σ_B .
 3. If all FDs in Σ_A are implied by Σ_B , then Σ_A is implied by Σ_B .
 4. Repeat the same process to check if Σ_B is implied by Σ_A . If both are implied, then Σ_A and Σ_B are equivalent.

Minimal Cover

- A set of FDs Σ is a minimal if:
 1. The **RHS** of every FD in Σ is a single attribute (i.e. FDs are in canonical form)
 2. The **LHS** of every FD in Σ is minimal (i.e. there are no extraneous attributes in the LHS)
 3. There are **no redundant** FDs in Σ (i.e. there is no FD in Σ that can be removed without changing the closure of Σ)
- Algorithm to compute minimal cover:
 1. Decompose FDs so that the RHS of every FD is a single attribute (i.e. put FDs in canonical form)
 2. For each FD, check if there are extraneous attributes in the LHS. If there are, remove them. To check if an attribute A in the LHS of an FD $X \rightarrow Y$ is extraneous, compute the attribute closure of $X - A$ under the set of FDs (including $X \rightarrow Y$). If the closure contains Y , then A is extraneous and can be removed from the LHS.
 3. For each FD, check if it is redundant. To check if an FD $X \rightarrow Y$ is redundant, compute the closure of X under the set of FDs without $X \rightarrow Y$. If the closure contains Y , then $X \rightarrow Y$ is redundant and can be removed from the set of FDs.
- Minimal cover is not unique. There can be multiple minimal covers for a given set of FDs.
- **Compact**: There is no FD with the same LHS.
- **Canonical Cover**: Compact and minimal cover.

The Chase Algorithm

- Test whether a given FD f is logically entailed by a set of FDs Σ .
- Construct a tableau with two rows and n columns, where n is the number of attributes in the relation schema.
- The first row is filled with α and the second row filled with distinct elements.
- For the FD $f : X \rightarrow Y$, replace the elements in the second row with α for all attributes in X .
- Apply the FDs in Σ to the tableau by changing distinct elements into α until no more changes can be made. Note: LHS column α .

Normalization